

Two's Complement:

1. Two's complement of 101010:

Step 1: Invert the bits

101010 $\rightarrow$ 010101

Step 2: Add 1

010101+1

=**010110**

2. Two's complement of 110011:

Step 1: Invert the bits

110011 $\rightarrow$ 001100

Step 2: Add 1

001100+1

= **001101**

3. Two's complement of 1001:

Step 1: Invert the bits

1001 $\rightarrow$ 0110

Step 2: Add 1

0110+1

=**0111**